

Qingyue Jiao

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U.S. Permanent Resident

PROFESSIONAL SUMMARY

Ph.D. candidate specializing in memory architectures, memory evaluation, and decision-making for AI agents. Experienced in designing multimodal memory systems, evaluation systems, and tool-use workflows to improve evidence-grounded reasoning and reliability across multi-turn tasks.

EDUCATION

University of Notre Dame

Ph.D. Candidate in Computer Science and Engineering; Advisor: Prof. Yiyu Shi

Notre Dame, IN

Sep 2024 – Present

Columbia University

MS in Computer Science, GPA: 4.0

New York, NY

Sep 2022 – Dec 2023

University of Michigan, Ann Arbor

B.S. in Computer Science & Physics (Double) GPA: 3.88

Ann Arbor, MI

Sep 2018 – May 2022

RESEARCH EXPERIENCE

University of Notre Dame

Ph.D. Candidate (Advisor: Prof. Yiyu Shi)

Notre Dame, IN

Sep 2024 – Present

MemEye: A Visual-Centric Evaluation Framework for Multimodal Agent Memory

- Designed a two-axis benchmark for **multimodal agent memory**, separating visual-detail retention from reasoning over changing states to enable targeted diagnosis of agent memory failures.
- Built a benchmark-construction pipeline with checks for answerability, visual dependence, and textual shortcuts; expanded coverage to 1,200 questions across eight scenarios and 12 evaluation categories.
- Evaluated 13 memory methods across four **vision-language model (VLM)** backbones, identifying lost visual details, stale retrieval, and state-tracking failures as bottlenecks in long-term agent reliability.

Hierarchical Memory for Radiology Report Generation with Rich Context

- Developed HM-RRG, combining **hierarchical memory** and **retrieval-augmented generation (RAG)** to ground chest X-ray reports in similar cases and longitudinal patient histories.
- Implemented image-conditioned memory compression, segment-level retrieval, and **LoRA-based parameter-efficient fine-tuning** of BioMistral-7B for selective use of long, noisy clinical context.
- Improved CheXbert clinical F1 from 0.483 to 0.580 (+20.1% relative) over flat context concatenation on MIMIC-CXR, demonstrating the benefit of structured memory for clinical accuracy.

Agent Last Doubt: Risk-Aware Tool Use and Decision-Making

- Developed a framework for **tool-using LLM agents** to clarify, search, act, or escalate under uncertainty, balancing expected task success, interaction cost, and irreversible error risk.
- Designed a modular, **action-conditioned value model** that predicts task success, cost, and critical errors separately, enabling configurable inference-time trade-offs without retraining the backbone agent.
- Built a **counterfactual rollout pipeline** on ToolSandbox and τ^2 -bench, testing alternative interventions from identical snapshots to generate outcome-based supervision for agent decisions.

Seeing, Maintaining, and Learning: The Evolution of Multimodal Agent Memory

- Co-organized an 80-author collaboration and co-developed a multimodal agent memory taxonomy spanning representations, **adaptive management**, and **self-evolving memory policies**.
- Curated an open-source reference covering 296 memory architectures and 83 evaluation resources across five modality categories, supporting systematic comparison of memory designs and benchmark coverage.
- Analyzed trade-offs in evidence preservation, storage efficiency, and temporal consistency, translating findings into design guidance for memory writing, retrieval, updating, and forgetting.

Node-as-Agent: Graph Agentic Network

- Developed ReaGAN, a **training-free, LLM-powered Node-as-Agent** framework enabling autonomous planning over local aggregation and global retrieval for node classification.
- Designed a **local-global memory** mechanism that combines graph-neighbor evidence with semantically relevant nodes retrieved through RAG, extending evidence access beyond direct graph connections.
- Evaluated node classification on Cora, Citeseer, and Chameleon with a frozen Qwen2.5-14B model; achieved 84.95% accuracy on Cora without task-specific fine-tuning.

MediQ-GAN: Medical Image Generation from Limited Data

- Implemented a prototype-conditioned **hybrid GAN** combining classical and quantum-inspired modules for **synthetic data augmentation** of underrepresented medical image classes.
- Designed and ran augmentation experiments across three datasets and two classifier backbones; improved ViT-small accuracy on ISIC2019 from 72.49% to 82.60% (+10.11 percentage points).
- Analyzed generator **Jacobian spectra and effective rank** to diagnose mode collapse, showing broader latent-space utilization than five classical GAN baselines.

PUBLICATIONS

- [1] **MemEye: A Visual-Centric Evaluation Framework for Multimodal Agent Memory**
Minghao Guo*, **Qingyue Jiao***, Zeru Shi*, et al.
International Conference on Learning Representations (ICLR 2027) (under review). arXiv:2605.15128.
- [2] **Hierarchical Memory for Radiology Report Generation with Rich Context**
Qingyue Jiao, Jiahao Zheng, Jun Zhuang, Yiyu Shi.
International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI 2026).
- [3] **Node-as-Agent: Graph Agentic Network**
Minghao Guo, Xi Zhu, **Qingyue Jiao**, Xiujin Liu, Haochen Xue, Chong Zhang, Shuhang Lin, Jingyuan Huang, Ziyi Ye, Yongfeng Zhang.
Workshop on Human-Centered Recommender Systems at The Web Conference (HCRS@TheWebConf 2026) (accepted). arXiv:2508.00429.
- [4] **Seeing, Maintaining, and Learning: The Evolution of Multi-modal Agent Memory**
Minghao Guo*, **Qingyue Jiao***, Xiujin Liu, et al.
Transactions on Machine Learning Research (TMLR 2026) (under review).
- [5] **OpenPM: Auditable Point-in-Time Evaluation for LLM Portfolio-Management Agents**
Xinying Cai*, Minghao Guo*, Jiahe Liu, **Qingyue Jiao**, Jiaojiao Han, Bangwei Guo, Yitao Long, Yuxuan Chen, Bohan Wu, Ruochen Wen, Wenxi Dai, Dimitris N. Metaxas, Raymond Li.
ACM SIGKDD Explorations 2026 (under review).
- [6] **Agentic Multimodal RAG: Roles, Decisions, and Evaluation**
Tianyu Yang, Wenliang Guo, Xiujin Liu, Shijian Deng, Minghao Guo, **Qingyue Jiao**, et al.
Transactions on Machine Learning Research (TMLR 2026) (under review).
- [7] **MediQ-GAN: Quantum-Inspired GAN for High Resolution Medical Image Generation**
Qingyue Jiao, Yongcan Tang, Jun Zhuang, Jason Cong, Yiyu Shi.
arXiv:2506.21015, 2025.
- [8] **A Comparison on Constrain Encoding Methods for Quantum Approximate Optimization Algorithm**
Yiwen Liu*, **Qingyue Jiao***, Yiyu Shi, Ke Wan, Shangjie Guo.
IEEE/ACM International Conference on Computer-Aided Design (ICCAD 2024).
- [9] **Synergizing Quantum Techniques with Machine Learning for Advancing Drug Discovery Challenge**
Zhiding Liang, Zichang He, Yue Sun, Dylan Herman, **Qingyue Jiao**, et al.
Scientific Reports, 14:31216, 2024.

*Equal contribution.

RELEVANT SKILLS

Machine Learning & AI: Agentic AI, Large Language Models (LLMs), Multimodal Large Language Models (MLLMs), Generative Modeling, Medical AI, Robot Learning and Control

Programming & Systems: Python, C++, C, Java, MATLAB, SQL, PyTorch, NumPy, SciPy

Software & Tooling: Linux, Git, HuggingFace, CUDA, Cloud & HPC Environments